Biochemistry lab

Experiment (8)

اصنع معروفاً واغرس فكرة

Dental Society

EXPIREMNT 8

Enzymatic colorimetric method (end point)

Purpose: to determine the concentration of components inside the serum by determining its absorbance, using spectrophotometer.

- Serum: plasma without RBC.
- Serum components: 1-Triglycerides (lipids)
  2-glucouse (carbohydrates)
  3-cholestrol (lipids)
  4-albumin (proteins)

Purpose of the experiment: measuring those components' concentrations.

***we will measure the concentration by beer's law:**

Remember: $A = \underbrace{a.b}_{\text{Constant}} .c \longleftrightarrow A1/C1 = A2/C2$

A: Absorbance.

C: Concentration.

a: absorption coefficient.

b: cuvette's width.

- First, here are some abbreviations you should know:

LPL: Lipoprotein lipase

ATP: Adenosine triphosphate

GK: Glycerol kinase

GPO: Glycerol phosphate oxidase

DHAP: Dihydroxy acetone phosphate

4-AA: 4-amino antipyrine

POD: peroxidase

FFA: Free fatty acid

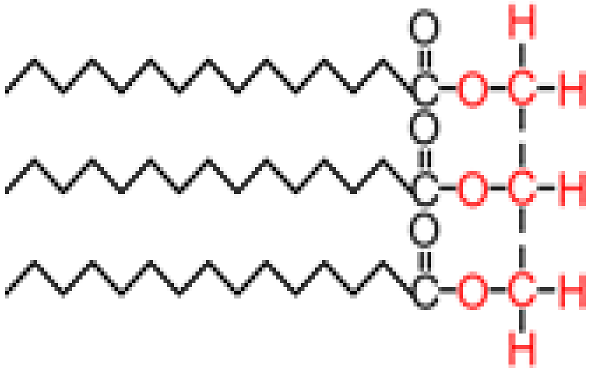
GOD: Glucose Oxidase

Very important

حفظهم مهم جدا.

1st experiment: measuring the concentration of triglycerides.

Triglycerides $\longrightarrow$



3 Fatty Acids + Glycerol

λ_{max} = 500 nm

- Triglycerides $\xrightarrow[\text{LPL}]{\text{Hydrolysis}}$ Glycerol + 3 FFA

- Glycerol + ATP $\xrightarrow[\text{GK}]{\text{phosphorylation}}$ Glycerol-3-phoshate + ADP

Those should be remembered.

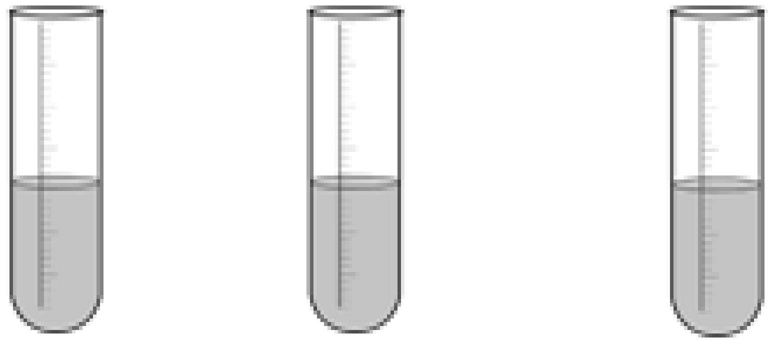
- Glycerol-3-phosphate + O_2 $\xrightarrow[\text{GPO}]{\text{oxidation}}$ DHAP + H_2O_2

- 4-AA +4 Phenol $\xrightarrow[\text{POD}]{H_2O_2}$ Quinoimine (PINK-RED) + H_2O

- as color's intensity increases, [H_2O_2] increases.
- [H_2O_2] describes the concentration of Triglyceride.
- Triglyceride's reagent consists of: **LPL, GK, GPO, POD**. When it reacts with triglyceride it gives us Quinoimine (pink – red)

- سلسلة المعادلات هذه هي للوصول للون ملحوظ. لذا، كي نصل لهذا اللون سنحتاج إلى مجموعة من العناصر. حفظهم مهم.
- تتناسب درجة اللون طرديا مع تركيز H_2O_2، تركيز H_2O_2 يعبر عن تركيز ال Triglyceride في العينة.

●Practical part:



1ml	1ml	1ml
(Blank)	10 µl	10 µl
Reset the device	Serum	standard (200mg/dl)

Remember: blank solution is the solvent without any solute; we reset the device on it to neglect the solvent's absorption.

"تماما مثل الميزان، نصفر الميزان بعد وضع الوعاء كي لا نحسب وزن الوعاء."

Procedure:

- Prepare three test tubes then add 1ml of reagent to each one of them.
- Add 10 µl of the sample (serum) to the second test tube.
- Add 10 µl of the standard to the third test tube.
- Measure the absorbance by spectrophotometer at 37 degrees at 500nm wavelength.
- Find the concentration of the sample by beer's law.

"نقوم بإضافة ال reagent على محلول ذا تركيز معروف من ال Triglyceride ونقوم بحساب A له، ثم نقوم بإضافة الReagent على Serum لنقوم بحساب A له، وبالقانون نحسب تركيز ال Triglyceride في الدم."

$A_1/C_1 = A_2/C_2$

A_1، A2 يتم اخذها من الجهاز مباشرة

C standard = 200 mg/dl

C sample = (A sample * C standard)/ A standard

2^{nd} experiment: Glucose

$$B-D-Glucose + H_2O + O_2 \xrightarrow{GOD} D-Gluconate + H_2O_2$$

$$\rightarrow \quad 4-AA + 4\,phenol \xrightarrow{H_2O\ +\ POD} Quinoneimine + H_2O$$

PINK-RED

Oxidation→ H_2O_2 appears → Quinoneimine appearance.

- Glucose's reagent consists of: **GOD, POD**. It gives Quinoneimine (pink).

- Practical part:

- Add 3 test tubes each with 1 ml of reagent
- Add of serum to the test tube number 2
- Add 10 µl of standard glucose to test tube number 3
- Because the first one is blank, contains the reagent only. Don't add anything else.
- Measure the absorbance at 37° C, λ = 500nm
- Then measure the concentration by beer's law.

λ_{max} = 500 nm

$A_1/C_1 = A_2/C_2$

A1 ,A2 يتم اخذها من الجهاز مباشرة

C standard = 100mg/dl

3rd Experiment: - Cholesterol

- Cholesterol is found in two shapes: Free (minor)

  Esterfied (major)

$$\text{Cholesterol ester} \xrightarrow{\text{Cholesterol esterase}} \text{Free Cholesterol}$$

$$\text{Free Cholesterol} \xrightarrow{\text{Cholesterol oxidase}} \text{Oxidized Cholesterol} + \underline{H_2O_2}$$

$$\text{4-AA} + \text{4phenol} \xrightarrow[\text{POD}]{H_2O} \text{Quinoneimine} + H2O$$

- Cholesterol's agent consists of: **Cholesterolesterase, POD**

"الشيء المشترك و الأساسي في الثلاث تجارب السابقة Triglyceride, Glucose, Cholesterol هي عملية الأكسدة Oxidization لتحصل على ال H_2O_2 الذي يظهر ال Quinoneimine."

- Practical part:-

- Add 3 test tubes with 1mL of reagent on each one.

- Test tube 1 will be Blank, add 10μL of the sample (serum) to test tube 2, and 10μL of standard to test tube 3.

- Then measure the absorbance at 37° and at

- Then measure the concentration by Beer's Law.

$\lambda_{max} = 500$ nm

$A_1/C_1 = A_2/C_2$

A1, A2 يتم اخذها من الجهاز مباشرة

C standard = 200mg/dl

4th Experiment: - proteins

• Albumin is considered 60 – 70% of plasma protein

$$\text{Albumin} \xrightarrow[\text{Acidic medium}]{\text{Bromocresol Green (BCG)}} \text{Green complex.}$$

• as Color's intensity increases, albumin's concentration increases

"درجة اللون اللي بتطلع تتناسب طرديا مع تركيز ال albumin."

•Albumin's reagent consists of: BCG.

• Practical Part:

- Add 3 test tubes with 1 ml reagent to each one
- Add 10 ml of the sample and of the standard to the test tubes
- Measure the absorbance , At 37 C with λ max = 630
- Find the concentration by beer's law.

λ_{max} = 630 nm

- Standard concentration for albumin = 5 mg/dl

→ Let's summarize it:

We have two ways to determine the concentration of the components in the plasma:

Enzymatic	Colorimetric
-Triglycerides - Cholesterol - Glucose	-Albumin
◄ هون بنضيف عليهم Enzyme مختلفين عشان يصير الهم لون ونقيس ل Absorbance و عن طريق Beers law نطلع ال concentration	◄ الفرق هون انه ما بضيف له Enzyme كمحفز بل استخدمنا dye (صبغة) عشان يطلع معنا complex ويتفاعل معه حسب ل Absorbance وال beers law

- **<u>Here we will try to summarize things you should remember:</u>**

Triglyceride's reagent: LPL, GK, GPO, POD

Triglyceride's λ_{max}: 500 nm

Triglyceride's standard conc.: 200 mg/dl

Glucose's reagent: GOD, POD

Glucose's λ_{max}: 500 nm

Glucose's standard conc.: 100 mg/dl

Cholesterol's reagent: Cholesterolesterase, POD.

Cholesterol's λ_{max}: 500 nm

Cholesterol's standard conc.: 200 mg/dl

Albumin's reagent: BCG (indicator)

Albumin's λ_{max}: 630 nm

Albumin's standard conc.: 5 mg/dl.

THE END

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Questions:

1-All of the followings are used as a reagent in triglycerides experiment except:

A – LPL

B – GOD

C – GK

D – POD

2- We add to all of the following experiments a different enzymes as a catalyst to get the concentration except?

A – Triglyceride.

B – Cholesterol.

C – Glucose.

D – Albumin.

3 – The colour of Quinoneimine is directly proportional to the concentration of H_2O_2?

A – True.

B – False.

Answers: -

1 – GOD (B)	2- Albumin (D)	3- True (A)